

Effective field theories of gravity 1

$$S = \int d^4x \sqrt{-g} \frac{M_{\text{Pl}}^2}{2} R$$

$$c = \hbar = 1$$

$$M_{\text{Pl}}^2 = \frac{1}{8\pi G}$$

$$R = g^{\mu\nu} R_{\mu\nu}$$

$$R_{\mu\nu} = R^\alpha{}_{\mu\alpha\nu}$$

$$S = \int d^4x \left[\frac{M_{\text{Pl}}^2}{2} R + \mathcal{L}_M \right] \sqrt{-g}$$

$\int d^4x \sqrt{-g}$: diffeo^m invariant measure
inv. under coordinate transf.

$$\int d^4x = \int \det \left[\frac{\partial x^\alpha}{\partial x'^\alpha} \right] d^4x'$$

$$\sqrt{-g} = \sqrt{-\det(g_{\mu\nu})} \rightarrow \frac{\partial x^\alpha}{\partial x'^\alpha} \sqrt{-g'}$$

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} (R - 2\Lambda) + \mathcal{L}_M \right]$$

$$\frac{\delta S}{\delta g_{\mu\nu}} = \frac{\delta}{\delta g_{\mu\nu}} \left(\sqrt{-g} g^{\mu\nu} R_{\mu\nu} \right) = -\sqrt{-g} G^{\mu\nu}$$

$$G^{\mu\nu} = R^{\mu\nu} - \frac{1}{2} g^{\mu\nu} R$$

$$\delta(\sqrt{-\det g} g^{-1}) \quad \delta g^{-1} = -g^{-1} \delta g g^{-1}$$

Λ, R leading terms in a derivative expansion
i.e. low energy effective field theory
for gravity

"Theorem" - Einstein GR unique low energy interacting theory for a massless spin-2 particle

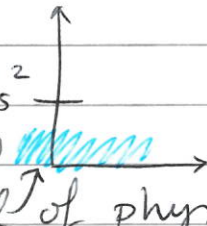
g weak coupling $\rightarrow$ small g

$$L_g = \frac{1}{g^2} [\dots]$$

sets weak coupling

String theory

$$\Lambda = m_s = \frac{1}{\sqrt{\alpha'}} \quad \begin{array}{l} \text{string length / mass} \\ \text{scale} \end{array}$$

for massless g spin 2 $g_{\mu\nu}$ $M^2 = 0$ 
scattering amplitudes

$$\mathcal{A}\left(\frac{E}{\Lambda}\right) = A_0(E) \left(1 + \# \frac{E^2}{\Lambda^2} + \# \frac{E^4}{\Lambda^4} + \dots \right)$$

α' expansion

what L would reproduce a similar $\nearrow$
we would need to add local terms like before
in order of $\left(\frac{1}{\Lambda}\right)$ ~~times~~ an extra $\frac{1}{g^2}$ in front

$$S_{\text{string}}^{10\text{dim}} \int d^{10}x \left[\frac{\Lambda^8}{g^2} \left[R + \alpha' R^2 \dots + \alpha'^2 R^3 \dots + \alpha'^3 R^4 \dots \right] \right]$$

($C=0$)

$$M_{\text{pl}}^8 = \frac{M_s^8}{g_s^2}$$

$M_{\text{pl}} \gg M_s$ corrections from α' kick in before M_{pl}

EFT of Gravity 2

Quantizing GR

recall: Maxwell $\sim$ covariant $\sim \partial^\mu A_\mu = 0$

add gauge fixing term

$$S_{gf} = \int d^d x \left[\frac{1}{2\xi} (\partial_\mu A^\mu)^2 \right] \quad \text{Feyn. gauge } \xi = 1$$

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}^2 - \frac{1}{2} (\partial_\mu A^\mu)^2 = -\frac{1}{2} \partial_\mu A_\nu \partial^\mu A^\nu = \frac{1}{2} \partial_\mu A_\nu \square A^\mu$$

Gravitons propagating in Minkowski

$$g_{\mu\nu} = \eta_{\mu\nu} + \frac{2}{M_{pl}} h_{\mu\nu}$$

$$S = \int d^d x \frac{1}{2} M_{pl}^2 \sqrt{-g} R = \int d^d x \left\{ -\frac{1}{2} \partial_\alpha h_{\mu\nu} \partial^\alpha h^{\mu\nu} + \frac{1}{2} \partial_\mu h^\mu{}_\nu \partial^\nu h^\mu{}_\mu - \frac{1}{2} \partial_\mu h^\mu{}_\nu \partial^\nu h^\mu{}_\mu \right\}$$

gauge choice: de Donder (harmonic gauge)

$$\partial^\mu \left(h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h \right) = 0$$

$$x^\mu \rightarrow x^\mu + \epsilon^\mu, \quad h_{\mu\nu} \rightarrow h_{\mu\nu} + \partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu$$

$$\Rightarrow \partial^\mu \left(h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h \right) \rightarrow \partial^\mu \left(h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h \right) + \square \epsilon_\nu$$

$$S = S_{GR}^{free} + \int d^d x \left\{ \frac{1}{2} \left(\partial_\mu h_{\nu\rho} - \frac{1}{2} \partial_\nu h_{\mu\rho} \right)^2 \right\} + S_M =$$

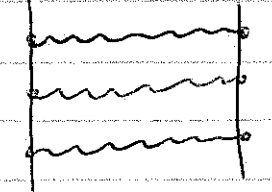
$$= \int d^d x \frac{1}{2} h_{\mu\nu} \square \left(h^{\mu\nu} - \frac{1}{2} \eta^{\mu\nu} h \right) + \frac{h_{\mu\nu} T^{\mu\nu}}{M_{pl}} + \text{non linear GR interactions}$$

$$\rightarrow \langle 0 | \hat{T} \exp \left\{ i \int d^d x \frac{h_{\mu\nu} T^{\mu\nu}}{M_{pl}} \right\} | 0 \rangle$$

$$= \int \mathcal{D} h_{\mu\nu} \exp \left\{ i \int d^d x \frac{1}{2} h_{\mu\nu} \square \left(-h^{\mu\nu} - \frac{1}{2} \eta^{\mu\nu} h \right) + \frac{h_{\mu\nu} T^{\mu\nu}}{M_{pl}} \right\}$$

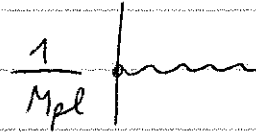
$$Z = \int \mathcal{D}g_{\mu\nu} \delta(F^M) \det \left[\frac{\delta F^M}{\delta \varepsilon^\nu} \right] e^{iS_{GR}} \quad \text{Fadeev-Popov}$$

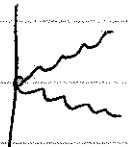
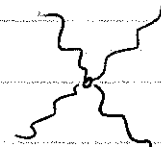
$$= \int \mathcal{D}g_{\mu\nu} \int \mathcal{D}\lambda_\mu \exp \left\{ i \int d^d x \lambda_\mu F^\mu \right\} \times \\ \times \exp \left\{ i \int d^d x \sqrt{-g} \left[\bar{c}^\mu \square c_\mu - \bar{c}^\mu c^\nu \bar{R}_{\mu\nu} \right] \right\}$$

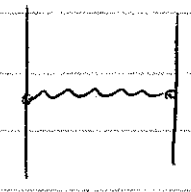
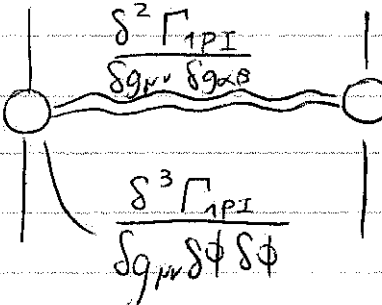
Example: 2 masses:  ladder diagram
graviton exchanged
multiple possibilities

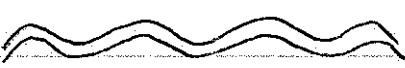
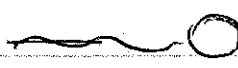
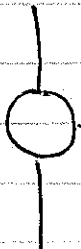
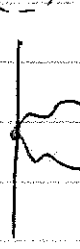
$$S = \int d^4 x \sqrt{-g} \left\{ \frac{1}{2} M_{Pl}^2 R - \frac{1}{2} (\nabla_\mu \phi)^2 - \frac{1}{2} M^2 \phi^2 \right\}$$

$$g_{\mu\nu} = \eta_{\mu\nu} + \frac{2h_{\mu\nu}}{M_{Pl}}$$

cubic $\rightarrow \frac{h_{\mu\nu} T^{\mu\nu}(\phi)}{M_{Pl}}$ 

$\rightarrow \frac{1}{M_{Pl}^2} h^2 T$  $\frac{1}{M_{Pl}}$ 

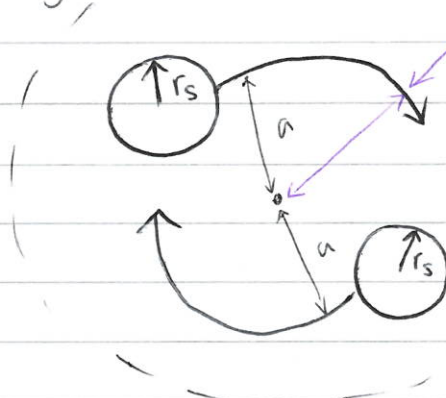
$\rightarrow$  $\Rightarrow$ 
Labels: $\frac{\delta^2 \Gamma_{1PI}}{\delta g_{\mu\nu} \delta g_{\alpha\beta}}$ and $\frac{\delta^3 \Gamma_{1PI}}{\delta g_{\mu\nu} \delta \phi \delta \phi}$

 =  + 
+  + 
 =  + 

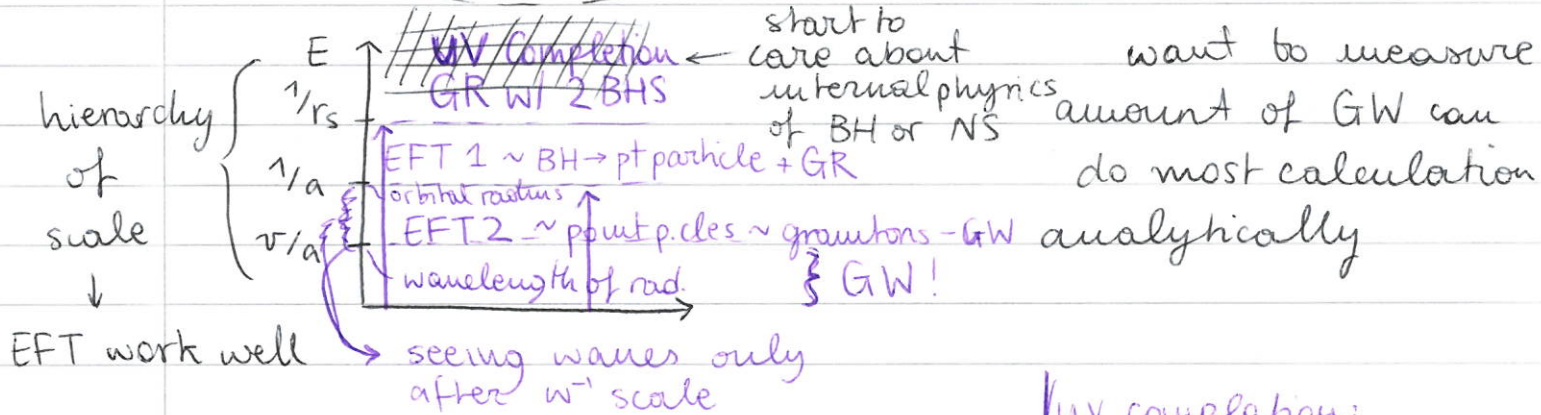
EFT of Gravity

EFT of Black Hole mergers

mostly classical but we use loops



rotating at: angular freq
 $\omega^{-1} \sim \frac{v}{a}$
 other $\omega a \sim v$
 length scale in system
 $\frac{1}{\omega} \sim \frac{v}{a}$



EFT 1: from 0 to $1/r_s \rightarrow$ Pt particle + GR [theory @ all energy]

EFT 2: = 0 to $1/a \rightarrow$ we don't see the binary system we see a blob

$\rightarrow$ Non-rel EFT bc valid only if $v \ll 1$

NR EFT matches w/ Post Newtonian expansion of GR

Black Hole: far away looks like a pt pale

- rather than having BH source in EFT1 we have point p.cle

EFT: 1 • Dof : Spin 2: $g_{\mu\nu}$, pt particle $\chi^\mu(x)$

• Symmetries: ST Diffeo^m

λ reparameterization 1-dim al

$\hookrightarrow$ like on a world diffeo^m sheet in string

lowest ~~order~~ orders
 curvature coupling to matter $\mathcal{L}$

lowest order interactions are not interesting
 → Redundant operators which can be removed by field redefs so we don't need to track them

$$\mathcal{L}_{\text{worldline}} = \dots + C_E \cancel{E_{\mu\nu} E^{\mu\nu}} + C_B B_{\mu\nu} B^{\mu\nu}$$

quasi electric field

leading order of terms that can't be redefined.
 Projections: quasi magnetic field

$$E_{\mu\nu} = R_{\mu\nu\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau}$$

dual Riem.

$$B_{\mu\nu} = \epsilon_{\mu\nu}{}^{\alpha\beta} R_{\alpha\beta\gamma\delta} \frac{dx^\gamma}{d\tau} \frac{dx^\delta}{d\tau}$$

Riem. tensor projected on a particle trajectory

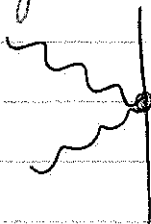
All of the physics of the BH is contained in coefficients C_E and C_B

UV completion theory: GR + BH

so we can compare the terms by showing that

$E \rightarrow 2$ gravitons and a particle
 compute scat. ampl $\mathcal{A}$

EFT 1



to infer vertex ~~you infer~~ Riem. $R \sim \partial\partial h$
 $\mathcal{L} \sim (\partial\partial h \otimes \dot{x}^\mu \dot{x}^\nu)^2$

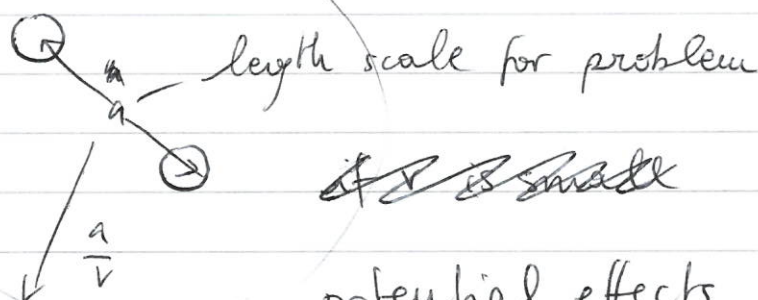
on the pt p.cle side

UV compl and on the BH side (scattering of GW from BH)
 sending coherent GW to BH

EFT 2 better to do it w/ Path Integrals

2 types of gravitational dynamics:

- potential effects: quasi static (extreme off shell virtual effects)
- radiation (on shell effect)



~~if λ is small~~

potential effects and radiation don't start at same scale

Scalar fields gen. funct. $W[J] = \frac{1}{2} \int J \frac{1}{\square + m^2 + i\epsilon} J d^4x$

$$e^{iW[J]} = \langle 0 | T e^{i \int d^4x J(x) \phi(x)} | 0 \rangle$$

Vacuum persistence probability =

$$= | \langle 0 | T e^{i \int d^4x J(x) \phi(x)} | 0 \rangle |^2$$

$W[J] \notin \mathbb{R}$ because which causes vacuum to decay

$$\propto e^{-2 \text{Im}[W(J)]}$$

Real part $\rightarrow$ potential eff.

Imag part $\rightarrow$ radiation part: which is why vacuum is decaying by producing particles

like dynamics of solar system

if system is slow bc ~~the~~

$$P\left[\frac{1}{k^2+m^2}\right] = P\left(\frac{1}{-k_0^2 + \vec{k}^2 + m^2}\right)$$

$k^0 \sim v k \rightarrow \partial_t$ are suppressed by velocity relative to ∂_x

obs not for radiative bc $\partial_t \sim \partial_x \rightarrow k^0 \sim k$

~~potential~~

if $k^0 \ll |\vec{k}|$ then $\frac{1}{-k_0^2 + \vec{k}^2 + m^2} \approx \frac{1}{\vec{k}^2 + m^2} + \frac{k_0^2}{|\vec{k}^2 + m^2|^2} + \dots$
 only valid Taylor exp in non relativistic

back in real space in $m=0 \sim \frac{1}{\vec{k}^2} + \frac{k_0^2}{|\vec{k}|^4} + \dots$

$$\int d^4x e^{ik(x-x')} \frac{1}{-k_0^2 + \vec{k}^2 + m^2} \approx \delta(t-t') \frac{1}{4\pi |\vec{x}-\vec{x}'|} - \frac{\partial^2 \delta(t-t')}{\partial t^2} \frac{|\vec{x}-\vec{x}'|^2}{2} + \dots$$

$\uparrow$ Yukawa version $\uparrow$ massless case $\downarrow$ local

What happens if you do that to something ~~the~~
 w/ $T^{\mu\nu}$?

$$\text{Re}[W[T^{\mu\nu}]] = \int d^4x \int d^4y T^{\mu\nu}(x) \left(T^{\mu\nu}(y) - \frac{1}{2} T_{\gamma\gamma}(y) \right) \times \\ \times \text{Re}[G(x-y)]$$

in potential approx $\rightarrow$ is satisfied by expansion that looks like

EFT 2 $\rightarrow$ couplings between BH 1 and 2

$$S_{\text{EFT2}} = \int dt \left[\sum_I \frac{1}{2} m_I \dot{\vec{r}}_I^2 - \right.$$

$$\left. - \sum_{I,J} \frac{m_I m_J}{M_{\text{pl}}^2 |\vec{x}_I - \vec{x}_J|} - \frac{1}{2} h_{00} - \cancel{h_{0i}} h_{0i} \dot{V}_I^i - \right.$$

$$\left. - \frac{1}{4} h_{00} \dot{\vec{r}}_I^2 - \frac{1}{2} h_{ij} \dot{V}_I^i \dot{V}_I^j \right.$$

$$\left. + \frac{1}{8} \sum_I \dot{\vec{r}}_I^4 + \frac{m_1 m_2}{|\vec{x}_1 - \vec{x}_2|} \left(3(\vec{x}_1^2 + \vec{x}_2^2) - 7\vec{x}_1 \cdot \vec{x}_2 - \right. \right.$$

$$\left. - \frac{(\vec{v}_1 \cdot \vec{x}_{12})(\vec{v}_2 \cdot \vec{x}_{12})}{|\vec{x}_1 - \vec{x}_2|} \right) + \dots$$



interactions between worldlines

~~the~~ method was done a long time ago
Einstein - Infeld-Hoffman Lagrangian

EFT Approach to Gravitational dynamics.